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Synthetic extremophiles via species-specific formulations improve microbial therapeutics

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Microorganisms typically used to produce food and pharmaceuticals are now being explored as medicines and agricultural supplements. However, maintaining high viability from manufacturing until use remains an important challenge, requiring sophisticated cold chains and packaging. Here we report synthetic extremophiles of industrially relevant gram-negative bacteria (*Escherichia coli* Nissle 1917, *Ensifer meliloti*), gram-positive bacteria (*Lactobacillus plantarum*) and yeast (*Saccharomyces boulardii*). We develop a high-throughput pipeline to define species-specific materials that enable survival through drying, elevated temperatures, organic solvents and ionizing radiation. Using this pipeline, we enhance the stability of *E. coli* Nissle 1917 by more than four orders of magnitude over commercial formulations and demonstrate its capacity to remain viable while undergoing tableting and pharmaceutical processing. We further show, in live animals and plants, that synthetic extremophiles remain functional against enteric pathogens and as nitrogen-fixing plant supplements even after exposure to elevated temperatures. This synthetic, material-based stabilization enhances our capacity to apply microorganisms in extreme environments on Earth and potentially during exploratory space travel.

Microorganisms have been central to human technological progress and continue to be key in wide-ranging fields from food production (for example, baked goods) to biologics manufacturing (for example, synthetic insulin)[1]. In these applications, the microbial cells are kept alive only during the manufacturing process and are destroyed, deactivated or removed from the final product. However, the pharmaceutical, agricultural and space health fields have now turned to developing live microorganisms as the final product to cure disease[2], to enhance crop production[3] and for on-demand bioproduction[4].

Critical to these new microbial technologies is the maintenance of high cell viability throughout the entire life cycle of the product. Currently this is achieved through well-established microbial cryopreservation methods[5] or with organisms that have natural extremophilic properties[6]. While acceptable for some initial applications, the need for select organisms or costly, inflexible cold chains severely limits the use case of live microbial products.

An ideal solution would be dry, shelf-stable microbial materials that are simple to package, ship and use. While extensive research has

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been done on the lyopreservation of microorganisms, these previous studies mainly focus on storage in culture collections that require only minimum viability for recovery through culture[7,8]. Furthermore, while general stabilizers are known, these studies have also shown that specific viability results vary widely depending on the microorganism in question[7,8].

Therefore, in this work, we first surveyed the current state of microbial stabilization in commercial products (that is, probiotics). We found generally low viabilities with particularly poor results for gram-negative bacteria (for example, *E. coli*). To fill this gap, we developed material-based synthetic extremophiles that are shelf stable without refrigeration and survive the extreme conditions in pharmaceutical manufacturing pipelines (for example, heat, pressure and solvents). Finally, we demonstrate that this stabilization approach maintains the functional capacity of the microorganisms against enteric pathogens, in a plant root nodulation assay and in ionizing radiation.

Survey of commercial probiotic products

A salient example of dried microbial products are the current commercially available probiotics marketed for human use that line pharmacy shelves. These products represent the current commercial standard of microbial stabilization. However, when we surveyed the viable cell counts (colony-forming units, CFUs) across a range of probiotics (Supplementary Table 1), we found only 7 in 13 products contained viable cell counts at or higher than the promised amount on the label (Fig. 1 and Supplementary Fig. 1), with a mean (geometric) viability of ~21% of that promised. Nevertheless, when we assessed the total cell count microscopically (both dead and alive cells), all products contained cell totals above those promised (Supplementary Fig. 1a), suggesting a loss of viability during and following manufacture. When we compared the viable cell count to this total cell count, only 1 in 13 products had a viability greater than 40%, with 6 in 13 products with viabilities lower than 2% and an overall mean viability of just 1.9% of total cells (Fig. 1).

We also found a lack of variety in terms of microbial species identity (Supplementary Fig. 1b). A majority (9 of 13) of products used microorganisms from just two groups (*Lactobacillus* and *Bifidobacterium*), which encompassed all the products with higher viabilities (Supplementary Fig. 1b). There was only one commercially available representative in the large clade of gram-negative bacteria (Mutaflor, *E. coli* Nissle 1917) and it had the lowest viability (0.05% of promised and 0.01% of total cells). To corroborate this finding, we found one additional commercial gram-negative product marketed for plant use (Nitragin Gold, *E. meliloti*) and found it also had a low viability (10% of promised and 1% of total cells; Fig. 1).

Furthermore, to evaluate whether these products had intrinsic extremophilic properties, we stress-tested at 50 °C for 24 h. We found nearly all the products retained an acceptable viability of >10% relative to non-stressed samples (Supplementary Fig. 2). However, the *E. coli* Nissle 1917 product Mutaflor retained only 0.002% viability (Supplementary Fig. 2). Overall, the above survey points to an industry consolidated around a small number of gram-positive microorganisms that are easy to stabilize and poor options for gram-negative organisms like *E. coli* Nissle 1917. By contrast, the synthetic biology and microbial therapeutics research fields in academia are centred on gram-negative bacteria with many of the most forward-looking examples (for example, cancer treatment) specifically using *E. coli* Nissle 1917 (refs. 2,9,10).

High-throughput pipeline for synthetic extremophiles

To overcome the incongruity between the commercial and academic fields, we developed a high-throughput pipeline to generate synthetic extremophiles (Fig. 2a and Methods). We defined a synthetic extremophile as a material–microbial formulation that has an enhanced

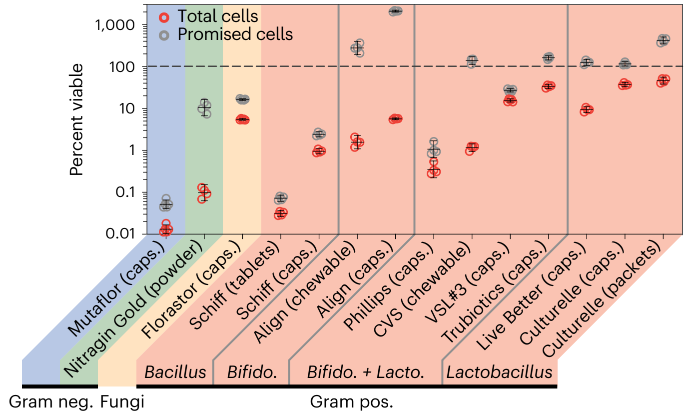

Fig. 1 | Survey of commercially available dry formulations of microbial probiotics. We quantified the viability of a wide representative range of commercial probiotic products (Supplementary Table 1). Viable cell counts were defined as the number of CFUs assessed through dilution plating on the appropriate solid medium and quantified via programmatic image analysis (Methods). Percent viable rates relative to promised cells were computed by dividing the CFU count by the CFU counts printed on the product label. Percent viable rates relative to total cells were computed by dividing the CFU counts by the number of total countable cells determined via automated fluorescence cytometry. $N = 4$ or 5 independent dosage forms. Geometric means and geometric 95% confidence intervals are plotted in black over individual replicates. Broad phylogenetic classes of the component microorganisms are noted in colour and specific genera are noted in italics (Supplementary Fig. 1 for detailed make-up). neg., negative; pos., positive; *Bifido.*, *Bifidobacterium*; *Lacto.*, *Lactobacillus*; caps., capsules.

capacity to survive abuse such as drying, heating and exposures to pressure, acid, organic solvents and ionizing radiation as compared to the original microbial strain.

We hypothesized that a synthetic extremophile that could survive in the dry state could then be enhanced to survive additional environmental extremes using established material formulation techniques applied to the dry powder (for example, a coating)[11]. In designing our approach, our overarching requirement was regulatory and industrial translatability. Therefore, we chose to apply material stabilizers rather than genetic changes that would add regulatory burden. Similarly, rather than develop new synthetic materials, we designed a material library (Supplementary Table 2) composed primarily of materials generally recognized as safe (abbreviated 'GRAS') by the Food and Drug Administration (FDA). With this limited set of compounds, we chose a highly parallelized method to map the material–organism viability space in contrast to previous piecemeal approaches. To accomplish this, we chose to dry the mixtures by freeze-drying in a shelf lyophilizer because it could be made compatible with liquid automation (Fig. 2a), as opposed to alternative drying methods such as a fluidized bed or spray drying.

We applied this pipeline to four microorganisms (*E. coli* Nissle 1917, *E. meliloti*, *S. boulardii* and *L. plantarum*) that span a wide range of phylogenetic backgrounds, are technologically important and are genetically tractable. Each microorganism was mixed with each of the 260 materials at two different concentrations to give 2,080 microbial–material formulations and then dried in batch. One material concentration was chosen to be close to the solubility limit of the respective material (5X; Supplementary Table 2) to ensure matrix formation in most materials, and the second dilution (1X; Supplementary Table 2) was chosen to account for possible inhibition of growth at high material concentrations during rehydration. Ultimately, lower concentrations proved to be most effective (vide infra). We chose an initial mild storage stress of 24 h at room temperature to differentiate microbial–material combinations by their thermotolerance. This contrasts with current

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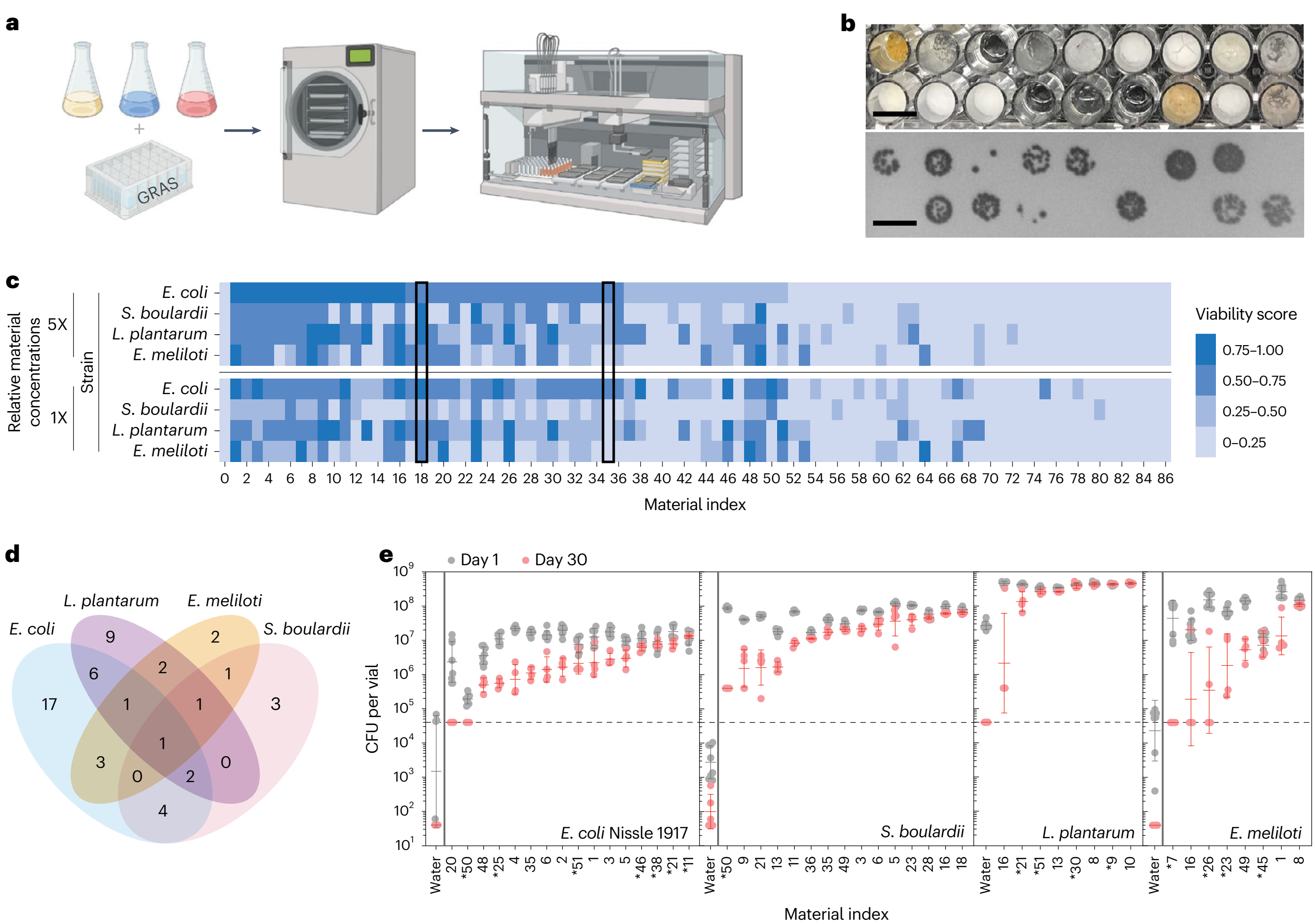

Fig. 2 | Development and validation of high-throughput pipeline for dry stabilized microbial materials. a, We developed a high-throughput pipeline (batch mixing, freeze-drying, plating and quantification) to assess the capacity of a library of materials generally recognized as safe (GRAS) to stabilize microorganisms (Methods). **b**, Representative dried formulations made in high-throughput pipeline and corresponding colony counting images. Scale bars, 1 cm. **c**, Result of materials with the top 33%, showing the superior viability of some materials relative to others after drying and storage for 24 h at room temperature; results are across four organisms and two concentrations and are ordered by descending capacity to stabilize *E. coli* Nissle 1917 at the 5X concentration. The viability score is a composite colony count normalized to the maximum observable growth defined as 1 (Methods). Supplementary Table 2 contains material identities and concentrations corresponding to the noted material indices, and Supplementary Fig. 4 contains full results. Black boxes mark the known lyoprotectants ATCC reagent 20 (index 18) and trehalose (index 35). $N = 3$ independent biological replicates. **d**, Correlation analysis of top-performing materials for each organism. **e**, Verification of top materials defined in **c** via a more precise assay (freeze-drying in vials and individual CFU counts on plates) and the ability to preserve viability for 1 and 30 days at room temperature. Material indices marked with asterisks are at 1X concentration, while others are at 5X. Horizontal dashed line represents the limit of detection. Where noted by a vertical bar, the water sample was assayed at higher concentrations to achieve a lower limit of detection. $N = 8$ independently dried vials for all day 1 groups. $N = 4$ independently dried vials for day 30 *E. coli* Nissle 1917, *E. meliloti* and *L. plantarum* groups. $N = 5$ independently dried vials for day 30 *S. boulardii* groups. Geometric mean and s.d. plotted over individual replicates.

microbial freeze-drying practice, which still requires refrigeration to maintain acceptable viability in the dry state[7].

Following this stress period, the residual viability of each formulation was measured via colony counts assessed at high throughput, resulting in a normalized viability score (Fig. 2b,c, Supplementary Fig. 3 and Methods). As expected, the included negative controls (that is, antimicrobial compounds sodium metabisulfite and sodium hydroxide) consistently gave viability scores of 0, and the positive controls (that is, known lyoprotectants trehalose and American Type Culture Collection (ATCC) reagent 20) consistently resulted in measurable viability (Fig. 2c and Supplementary Fig. 4, black boxes). Additionally, as expected from the lyoprotectant literature[7], sugars and peptones were overrepresented in top-performing formulations (Supplementary Fig. 5). Surprisingly, neither the positive controls nor any of the single materials were top performers across the range of organisms. Instead, our results show that ideal species-specific formulations exist, with low overlap in best-performing materials even between the two gram-negative representatives (that is, *E. coli* Nissle 1917 and *E. meliloti*; Fig. 2d).

Next, to further stratify the top-performing materials for each of the four microorganisms, we increased the stringency of the storage stress by evaluating colony counts after extended storage for one month at room temperature (Fig. 2e). Stratifying in this way also provided a fingerprint for the intrinsic extremophilic qualities of the chosen organisms. For example, our results indicate that the lactic acid bacteria *L. plantarum* already had good thermotolerance, as nearly all the top-performing materials were indistinguishable after storage (Fig. 2e). Conversely, the two gram-negative bacteria showed a steeper drop-off with only a handful of materials maintaining the initial viability long-term. These results further corroborate our probiotic viability survey, suggesting that the lactic acid bacteria that dominate the market may be naturally more tolerant and easily stabilized by a wide range of materials.

Optimization of *E. coli* Nissle 1917 synthetic extremophile

Due to its importance in the microbial therapeutics and synthetic biology fields, we next chose to further enhance the extremophilic qualities of *E. coli* Nissle 1917. Our screen showed that previously established mixtures, such as the positive control ATCC reagent 18, tended to outperform the individual constituent components of the mixtures (that is, tryptic soy broth, sucrose and albumin for ATCC reagent 18). Therefore, we next compared two-material combinations of the library members with the best-performing material (melibiose) against the single-material formulations and evaluated the residual viability after 24 h at both room temperature and 50 °C (Fig. 3a). We defined top hits as material combinations with viability scores above the control (melibiose + vehicle). From these we excluded library members that were themselves mixtures (for example, lysogeny broth (LB)) or of animal origin (for example, mucin) to ease future manufacturing. The stability of the chosen two-material formulations (melibiose in combination with fructooligosaccharides, caffeine or yeast extract) was analysed in a larger format at 37 °C for extended storage times, and the mixtures with caffeine or yeast extract were found to be robust (Supplementary Fig. 6). We next titrated the relative concentrations of melibiose, caffeine and yeast extract to find the optimal formulations, which we termed formulations D and E (Fig. 3b).

Both melibiose and formulation D outperformed the commercial product Mutaflor (*E. coli* Nissle 1917) by over 3.5 orders of magnitude when stored at room temperature for one month (Fig. 3c). To confirm these results and exclude any manufacturing bias (that is, freshly made versus store-bought product), we also formulated *E. coli* Nissle 1917 in maltodextrin (the stabilizer used in Mutaflor) using our specific freeze-drying processes. We then compared this 'commercial-like' dry *E. coli* Nissle 1917 with our top-performing formulations. We found that at 37 °C, the maltodextrin-stabilized material lost all measurable viability (>4 orders of magnitude) within 11 days (Fig. 3d). By contrast, formulation D retained over 10% viability even after 6.5 months at 37 °C (Fig. 3d and Supplementary Fig. 7). This was greater than two orders of magnitude better than the unoptimized melibiose formulation. In addition to maintenance of relative viability, we also showed that the absolute CFU count could be optimized by altering the growth medium (Supplementary Fig. 8), time of harvest (Supplementary Fig. 9) and percent loading of bacteria (Supplementary Fig. 10) to produce synthetic extremophiles with viabilities of $>10^{10}$ CFU per gram (Supplementary Fig. 11).

To understand the mechanism of protection, we used transmission electron microscopy to compare the ultrastructure of *E. coli* Nissle 1917 cells dried in either the well-performing formulation D or the poorly performing commercial-like formulation (maltodextrin). The formulation D samples had a significantly lower fraction of cells with cell envelope defects (that is, detachment of the inner membrane from the outer membrane with the periplasmic space filled with electron-dense material; Fig. 3e and Supplementary Fig. 12). This difference between formulation D and maltodextrin was larger for samples exposed to 50 °C (Supplementary Fig. 12d). While protection against this membrane defect may arise via several mechanisms, one potential mechanism may involve the stabilization of membrane proteins critical for survival through the rehydration process. This hypothesis is supported by the observation that stationary-phase *pssA*-null *E. coli* mutants, which lack phosphatidylethanol-amine, leading to misfolded membrane proteins, also develop this specific membrane defect[12,13] (Supplementary Discussion).

Compatibility with pharmaceutical manufacturing methods

These synthetic extremophiles open the field of potential use cases for microbial products to treat disease, enhance agricultural output and support exploratory space travel (Fig. 4a). One immediate opportunity for human and animal health would be the ability to make dosage forms with tuned release parameters (for example, delayed release, enteric coatings and so on). However, this requires applying pharmaceutical manufacturing techniques normally too harsh for living cells such as milling (exposure to shear forces), wet granulation (exposure to isopropanol and baking), tableting (exposure to high pressure) and spray coating (exposure to acetone and baking). Therefore, we submitted the *E. coli* Nissle 1917 synthetic extremophile to this battery of processes (Fig. 4b) and characterized its robustness compared to the commercial-like alternative (maltodextrin; Fig. 4c). We found that the synthetic extremophile could be milled, mixed with excipients (that is, binder, glidant and filler) and tableted in open air at room temperature and still retain >15% viability, which was 2.4 orders of magnitude higher than the commercial control (0.06% viability; Fig. 4c). Wet granulation was similarly well tolerated (>28% viability; Fig. 4c). We also found that tableting pressures of up to 1 GPa were well tolerated (>15%), and storage at temperatures of up to 70 °C still had a half-life of two days (Supplementary Fig. 13a,b). We found that tablets were insensitive to the presence of oxygen, but sensitive to humidity with a half-life of 6 h at 80% relative humidity (Supplementary Fig. 13c,d).

Once tableted, the synthetic extremophile also survived spray coating with an acetone–isopropanol solution of the enteric polymer EUDRAGIT S 100 (viability, 0.5%), whereas the commercial-like alternative lost all measurable viability (Fig. 4c). We further optimized the coating process by reducing the post-coating drying temperature (Methods), leading to a 28-fold increase in viability to 14% relative to the freeze-dried powder (Fig. 4c, purple). Additionally, we determined that both the uncoated and enteric-coated tablets of *E. coli* Nissle 1917 passed quality control tests for friability, weight variation, tablet hardness and thickness (Supplementary Fig. 14a–e). The enteric-coated tablets also passed a compendial disintegration test in simulated gastric fluid and successfully protected the bacterial cells from the low pH environment (Fig. 4d and Supplementary Fig. 14f).

Tuning biosynthetic function through dosage form design

We next asked whether the synthetic extremophile could also remain functional in the tableted state and whether that function could be tuned through dosage form design. To test this, we incorporated a bioluminescent biosynthetic pathway (the luxCDABE operon) and measured luminescence upon rehydration. While the dry tablet showed no measurable luminescence, 48% of the ultimate luminescent level was reached within 15 min of rehydration (Fig. 4e). Substituting the tablet filler by the matrix-forming material hydroxypropyl methylcellulose (HPMC) led to a slow release of the bacteria and tuning of the kinetics of the luminescent function over a period of 24 h (Fig. 4e).

Synthetic extremophiles as microbial therapeutics

Next, we asked if the known capacity of *E. coli* Nissle 1917 to inhibit enteric pathogens was impacted by our formulation[14]. We first characterized the effect on *Shigella flexneri* 2a, the leading cause of diarrhoea-associated deaths in low-income and middle-income countries[15]. Specifically, we used a quantitative polymerase chain reaction (qPCR) co-culture inhibition assay after mild storage (4 °C, six weeks) of the synthetic extremophile (formulation D) and the commercial-like formulation (maltodextrin; Supplementary Fig. 15). Our results show that both fresh *E. coli* Nissle 1917 and its synthetic extremophile version (formulation D) inhibit *S. flexneri* by >99.8% in co-culture, while the commercial-like formulation showed no increase in effect compared to a non-inhibitory strain of *E. coli* (Supplementary Fig. 15b). Control experiments showed that stabilizers alone did not cause inhibition, nor did they modify the inhibitory effect when added to fresh *E. coli* Nissle 1917 cells (Supplementary Fig. 15c,d).

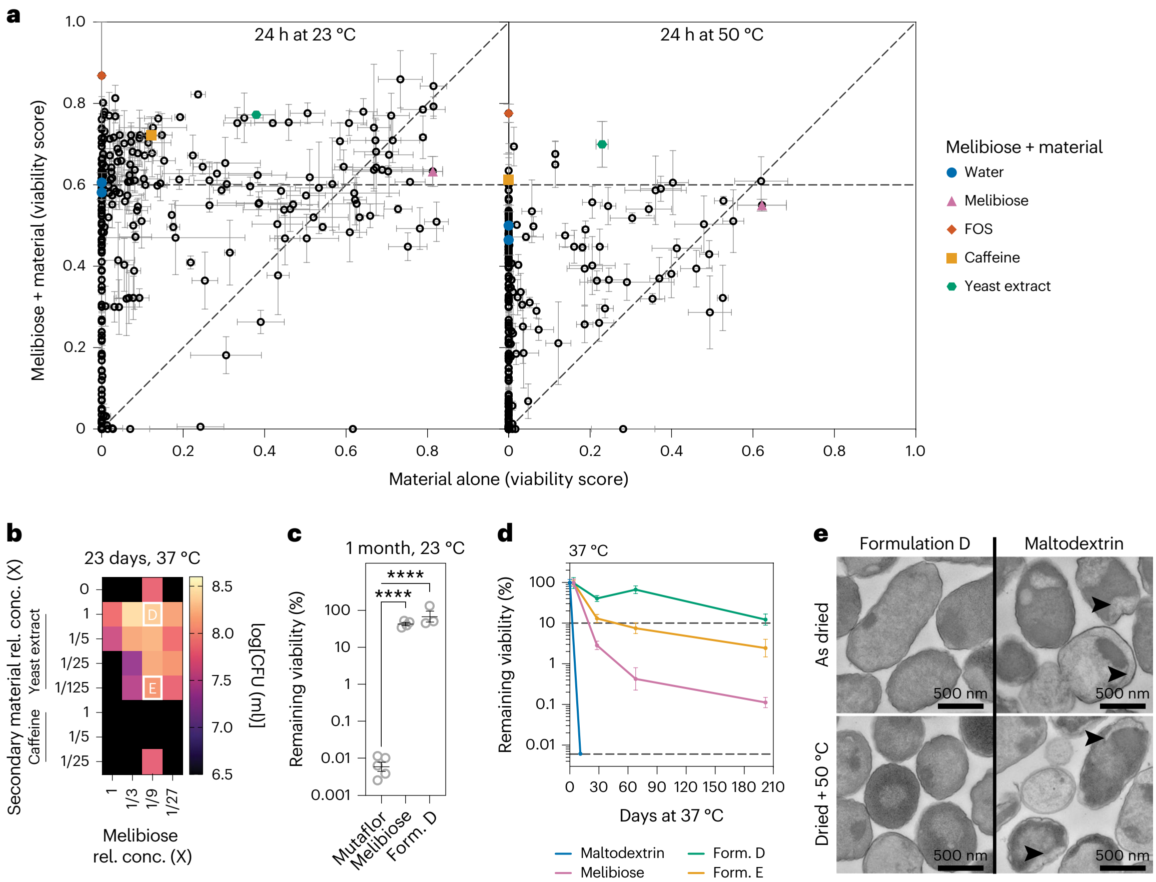

Fig. 3 | Application of pipeline to make *E. coli* Nissle 1917 into a synthetic extremophile. a, The library of materials was assayed individually (*x* axis) and in combination with melibiose (*y* axis) for each member's capacity to stabilize *E. coli* Nissle 1917 at room temperature and 50 °C for 24 h. Black circles mark the mean viability score (Methods) along both dimensions, and error bars represent the s.e.m. *N* = 3 independent biological replicates. Coloured shapes mark material combinations with melibiose selected for further characterization. Horizontal dashed line is a reference that marks the viability score of melibiose combined with vehicle (water, blue circle). Diagonal dashed line is the identity. All concentrations were at 1X as defined in Supplementary Table 2. FOS, fructooligosaccharides. **b**, The relative concentrations (rel. conc.) of each of the components in the selected two-material combinations (melibiose + yeast extract or melibiose + caffeine) were varied, and the resulting effect on viability was measured after storage at 37 °C for 23 days. X, standard concentration of each material as defined in Supplementary Table 2. White letters D and E mark the selected formulations for further characterization. *N* = 3 independent biological replicates. **c**, Direct stability comparison of formulation (Form.) D (as noted in **b**), the parent formulation (melibiose) and capsules of the commercial *E. coli* Nissle 1917 product Mutaflor after storage at room temperature for one month. Mean and s.e.m. of log values plotted over individual replicates. Independent biological replicates were used for each group: Mutaflor (*N* = 5), melibiose (*N* = 4) and formulation D (*N* = 3). Ordinary one-way analysis of variance (ANOVA) is as follows: Mutaflor–melibiose, ****$P < 0.0001$; Mutaflor–formulation D, ****$P < 0.0001$. **d**, Characterization of ultra-long stability at 37 °C compared to maltodextrin (stabilizer used in Mutaflor). Geometric mean and s.d. plotted for each time point. *N* = 3 independently measured samples from one large material batch. Lower dashed line is the limit of detection. Upper dashed line marks the 10% viability level. **e**, Transmission electron micrographs of *E. coli* Nissle 1917 after drying in formulation D or maltodextrin and optionally exposed to high temperature. Arrowheads mark detached inner membranes.

Then, to characterize the generality of these results and remove growth or PCR amplification bias, we carried out a more stringent test against *S. flexneri* 2a and two additional enteric pathogens, *Salmonella enterica* Infantis and *Shigella sonnei*, which are both emerging pathogens, associated with zoonosis during poultry production[16] and food-born illnesses in high-income countries[17], respectively. Specifically, we used a non-culture assay (phosphate-buffered saline (PBS) rather than LB, and 6 h rather than 10 h + outgrowth) quantified by direct plating, and increased storage condition stringency to 37 °C for five days (Fig. 5a). For this assay we directly compared the effect of the synthetic extremophile (formulation D) on pathogen load relative to the commercially purchased Mutaflor product. Our results show that formulation D leads to a significantly lower pathogen load in all three cases: 7-fold lower for *S. flexneri* 2a, 2.7-fold lower for *S. enterica* Infantis and 1.5-fold lower for *S. sonnei* (Fig. 5a). Control experiments showed that this effect requires the presence of viable *E. coli* Nissle 1917 cells (Supplementary Fig. 16).

To further evaluate the potential of formulation D as a microbial therapeutic, we carried out the same assay using a live-animal model of *S. flexneri* 2a intestinal infection in adult Yorkshire pigs (~60 kg; Fig. 5b). In this model, the pathogens and the therapy are independently injected into multiple, surgically accessed intestinal sections of a live animal in order to have internal controls and reduce animal usage (Methods). As before, the synthetic extremophile (formulation D) and the purchased Mutaflor product were exposed to harsh storage conditions (37 °C, five days) and then characterized for their effect on in vivo pathogen load through direct plating. Our results show that formulation D leads to a significantly lower (by sixfold) *S. flexneri* 2a load relative to Mutaflor (Fig. 5b) in pigs.

Synthetic extremophiles as agricultural supplements

Due to the potential of microbial nitrogen fixation for increasing agricultural sustainability with a lower need for chemical fertilizers,

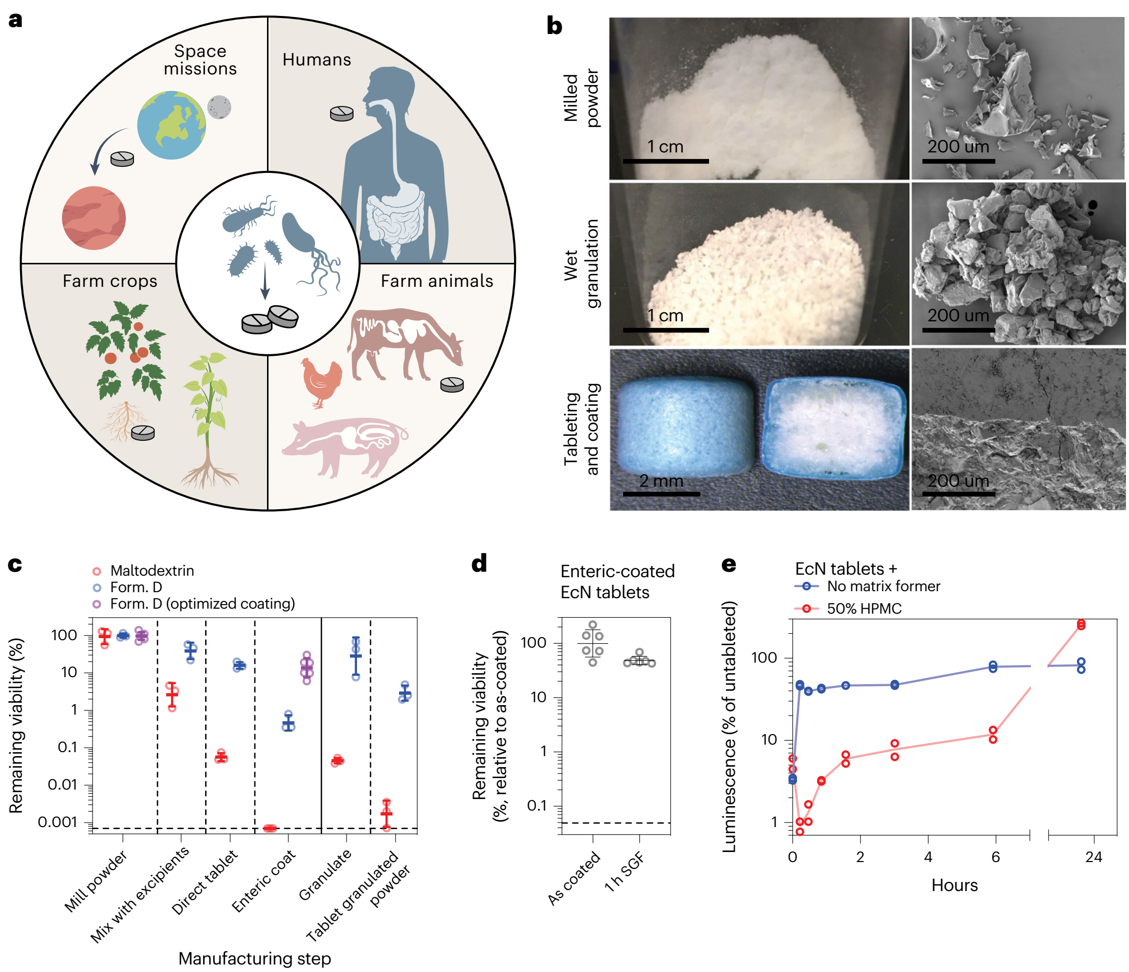

Fig. 4 | Synthetic extremophiles withstand harsh processing manufacturing processes. **a**, Microbial tableting and coating allow simple dosing, transport and tuning of release to access a wide range of applications. **b**, Photos and scanning electron micrographs of synthetic extremophiles processed via industrially relevant steps. **c**, Stability of the *E. coli* Nissle 1917 synthetic extremophile (Form. D) through each process relative to parent powder. Maltodextrin is the stabilizer in the commercial product Mutaflor. The optimized coating is described in the Methods. Dotted horizontal line shows the limit of detection. Geometric mean and s.d. plotted over individual replicates. $N = 3$ independent biological replicates. **d**, Tablets of the *E. coli* Nissle 1917 (EcN) synthetic extremophile withstand exposure to simulated gastric fluid (SGF, in compendial disintegration apparatus) when coated with the enteric polymer EUDRAGIT S 100. Geometric mean and s.d. plotted over individual replicates. $N = 6$ tablets. **e**, Tableting allows tuning of release kinetics through the inclusion of a matrix former (HPMC). Synthetic extremophile *E. coli* Nissle 1917 cells carry a biosynthetic pathway (the luxCDABE operon), and the released luminescence was tracked over time. Lines connect the means. Individual replicates plotted. $N = 2$ tablets.

we asked whether our approach to making the synthetic extremophiles would interfere with this complex symbiotic process in live plant roots[3,18]. This symbiotic process occurs between legumes and bacteria (rhizobia) in the soil via the formation of root nodules that house the bacteria that can then fix atmospheric nitrogen (N_2) into a usable form for the plant[18]. Specifically, we chose to evaluate nodule formation by the nitrogen-fixing bacteria *E. meliloti* in *Medicago truncatula*, a model legume used for studying this process[18,19]. *E. meliloti* is currently used as a seed inoculant prior to planting; however, a need remains for improved stability of the microbial product during the adverse conditions present during formulation, storage and seed inoculation, particularly when it comes to tolerance of elevated temperatures above 40 °C (refs. 20,21). Therefore, to generate a synthetic extremophile version of *E. meliloti*, we repeated the heat shock screen at 50 °C to find materials that would give *E. meliloti* synthetic thermotolerance (Supplementary Fig. 17). From these materials, we chose three optimal materials (sucrose, soytone and maltose) and two suboptimal materials (maltodextrin and polydextrose). *E. meliloti* was formulated in each, submitted to a 50 °C shock for 24 h, rehydrated and applied to seedlings of *M. truncatula* A17 (Fig. 5c). Functional symbiosis was measured as the percent of seedlings that successfully formed nodules. The synthetic *E. meliloti* extremophiles made with sucrose and maltose were able to successfully nodulate, while the one made from soytone led to plant stunting (Fig. 5c,d). These results show that our high-throughput pipeline can define a set of materials that are not just ideal for the target microbial species, but that can be further selected for those that are compatible with the specific target application, which may include complex biological processes such as nodulation.

Compatibility with ionizing radiation for space flight

Finally, to validate the compatibility of our synthetic extremophile approach with use in exploratory space flight, we characterized the hardiness of stabilized *E. coli* Nissle 1917 to a range of ionizing radiation doses from 100 to 10,000 Gy (typical radiation dose values are ~15 μGy per day on Earth[22], ~200 μGy per day on the Moon[23] and ~250 μGy per day on Mars[24]; a total accumulated dose for a 500 day trip to Mars is

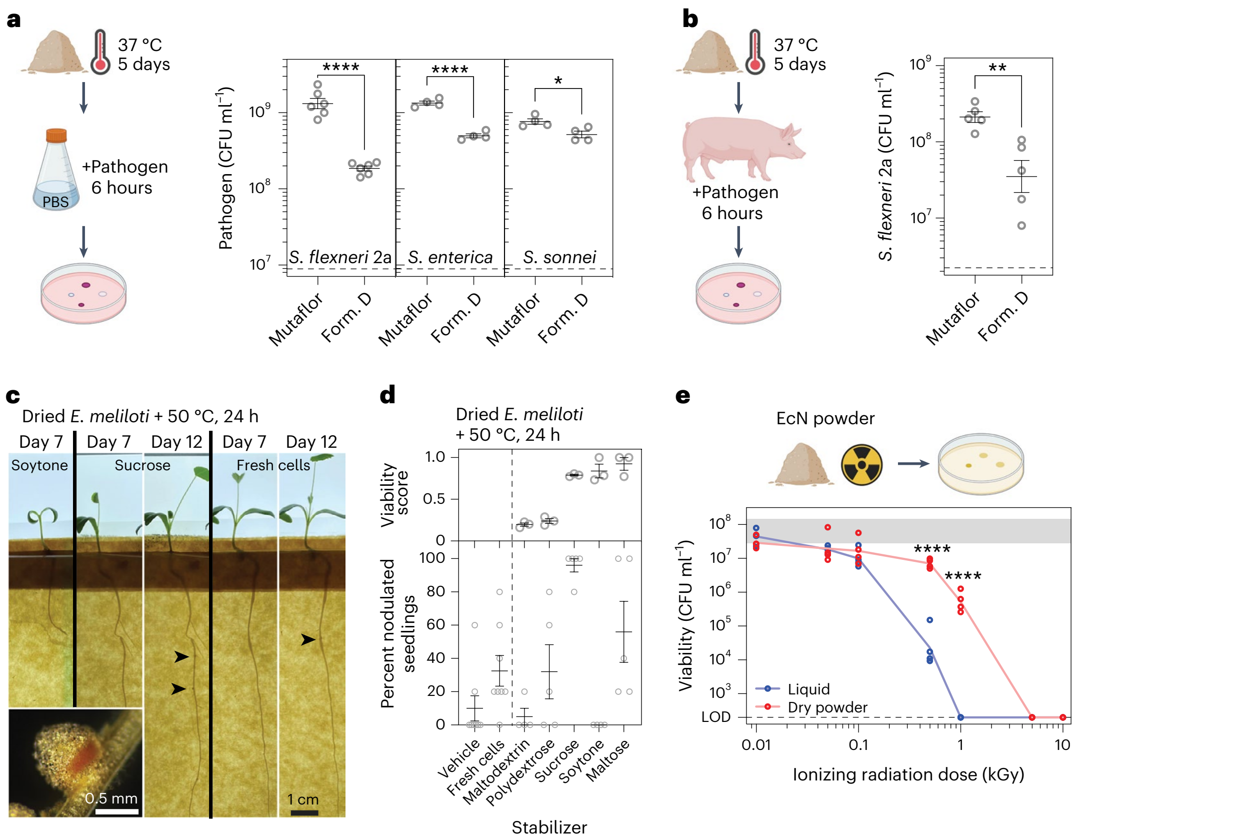

Fig. 5 | Synthetic extremophiles remain effective microbial therapeutics despite extreme temperature and radiation exposure. a, The *E. coli* Nissle 1917 synthetic extremophile (Form. D) is more effective than the commercial product Mutaflor against three enteric pathogens (*S. flexneri* 2a, *S. enterica* Infantis and *S. sonnei*) in vitro. Formulation D and Mutaflor were incubated at 37 °C for five days, rehydrated and mixed with the noted pathogen for 6 h in PBS. The pathogen load was quantified through differential plating on MacConkey agar and imaging (*E. coli*, stained pink, lactose fermentation positive (Lac+); pathogens, colourless, lactose fermentation negative (Lac−)). For each sub-panel of the plot on the right, the mean and s.e.m. of log values are plotted over individual replicates. Independent biological replicates were used for each group: *S. flexneri* 2a ($N = 6$), *S. enterica* Infantis ($N = 4$) and *S. sonnei* ($N = 4$). Dashed line is the limit of detection. Unpaired *t*-test, two-tailed *P* is as follows: *S. flexneri* 2a, $^{****}P < 0.0001$; *S. enterica* Infantis, $^{****}P < 0.0001$; *S. sonnei*, $^{*}P = 0.0228$. **b**, Formulation D is also more effective than Mutaflor at reducing *S. flexneri* 2a in a live-animal intestinal model of infection where pathogens are directly injected into intestinal segments. Formulation D and Mutaflor were processed as in **a**, but instead injected in the jejunum of live Yorkshire pigs and extracted after 6 h. Each replicate was quantified as in **a**. Mean and s.e.m. of log values plotted over individual replicates. $N = 5$ independent intestinal segments. Unpaired *t*-test, two-tailed $P = 0.0078$. **c**, *E. meliloti* is a nitrogen-fixing bacterium that supplies plants with nitrogen at symbiotic root nodules (inset). Synthetic extremophiles of *E. meliloti* were exposed to 50 °C, hydrated and tested for functionality in a root nodulation assay with *M. truncatula*. Nodulated seedlings (black arrowheads) can be quantified at day 12 post-root inoculation. **d**, Quantification of nodulation assay (**c**) compared to the material viability score. Mean and s.e.m. plotted over individual replicates. $N = 3$ independent biological replicates for the viability score groups. Nodulation percent was calculated in independent replicate seedling pouches for each group: controls left of dashed line ($N = 8$); maltose, sucrose, polydextrose ($N = 5$); maltodextrin, soytone ($N = 4$). **e**, EcN powder exposed to ionizing radiation is plated to quantify viability (CFU ml^{-1}). Synthetic extremophile *E. coli* Nissle 1917 cells are robust to ionizing radiation compared to a liquid suspension of cells. Lines connect the means. Individual replicates plotted. $N = 4$. Dotted line is the limit of detection (LOD). Grey shading indicates the viability range of sample not exposed to radiation. Two-way ANOVA is as follows: 0.5 kGy, $^{****}P < 0.0001$; 0.1 kGy, $^{****}P < 0.0001$.

0.32 Gy (ref. 25); and the dose for 444 days outside the International Space Station is 0.41 Gy (ref. 26)). The maximum value of 10 kGy represents the dose generally accepted for radiation-based sterilization of food[27]. We found that the synthetic extremophile could sustain exposures up to 1,000 Gy (Fig. 5e). At this radiation level a liquid suspension of the same bacteria lost all measurable viability.

Conclusions

We developed an approach for the identification of materials capable of imparting microorganisms with extreme environmental tolerance. This approach fills a gap in commercially available dry formulations (that is, probiotics), which we found tend to rely on a set of already resilient microorganisms. In particular, we applied our approach to increase the robustness of two gram-negative bacteria important for human health (*E. coli* Nissle 1917)[9] and agricultural productivity (*E. meliloti*)[19]. Specifically, we showed in vitro and in live animals and plants that these stabilized microbial powders function against enteric pathogens (*E. coli* Nissle 1917) and for the generation of root nodules for nitrogen fixation (*E. meliloti*) despite exposure to elevated temperatures. We also showed that these synthetic extremophiles withstand pharmaceutical manufacturing workflows enabling the production of controlled release microbial dosage forms (Supplementary Discussion).

Further enhancements in microbial stability and dosage form design could be achieved with approaches based on genetic mutations[28] or single-cell encapsulation techniques[29]. Our bulk material stabilization approach is orthogonal to and could be combined with these approaches to make microbial materials with more advanced robustness and pharmacokinetic release profiles.

In summary, these synthetic extremophiles stand to transform our capacity to disseminate bioactive organisms across human applications, from shelves across the globe to fields for agricultural practices to shuttles for space exploration.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41563-024-01937-6.

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Methods

Viability survey of commercial probiotic and microbial products

All products (Fig. 1 and Supplementary Fig. 1) were purchased and analysed in 2019 well before their expiration dates. The specific product names, lot numbers, expiration dates, dates of analysis and dosage forms are summarized in Supplementary Table 1. All procedures were analysed under sterile conditions. The dosage forms encountered were two-piece capsules filled with loose powder, sachets (packets) filled with loose powder or tablets. In each case only the microbial fraction was analysed. Specifically, for capsules, the two halves were carefully separated and the internal contents poured out onto weighing paper for further analysis. For sachets, the sachets were carefully torn open and the internal contents poured out onto weighing paper for further analysis. For tablets, the tablet was placed in a sterile ziplock bag and crushed by manually rolling a plastic tube over the tablet, and the resulting powder was poured out onto weighing paper for further analysis.

Promised CFU counts per dose were recorded from the product packaging and converted to CFU per gram by dividing by the mean mass of four doses. For Florastor, the amount of yeast cells was reported in milligrams per dose, and this number was converted to CFUs per dose by dividing by the approximate cell dry weight of yeast (~20 pg, 32% of 60 pg; BioNumbers ID (BNID) 101795 and BNID 105094; ref. 30). The dose mass includes only the recovered microbial fraction as described above (that is, internal contents of the dosage form). In cases that products report a CFU count both 'at manufacture' and 'by expiration date', only the CFU count 'by expiration date' was used. These data are summarized in Supplementary Table 1.

Viable CFU counts per dose were determined by rehydrating one dose in PBS (Thermo Fisher 10010049) at the specified rehydration ratio on ice; making tenfold serial dilutions in PBS on ice; plating 100 µl of each dilution onto the appropriate solid medium in 100-mm-diameter circle Petri dishes with 4.5 mm plating beads (Zymo); incubating at the appropriate conditions; and counting colonies on the dilution, which gave ~200–1,000 distinct colonies. Four or five separate doses of each product were prepared as described. As noted in Supplementary Table 1, the rehydration ratio was typically one dose in 50 ml, but in a few instances the rehydration volume was lowered when the cell counts were found to be low during an initial analysis. The medium and culture conditions used for each product were chosen based on the microorganisms listed on the package and are listed in Supplementary Table 1. Medium components were all purchased from BD Life Sciences as specified in Supplementary Table 3, solidified with 1.5% w/v Bacto Agar (BD, 214010) and prepared according to the label instructions. For aerobic conditions, plates were incubated in a static incubator, and for anaerobic conditions, plates were first placed in a BD GasPak EZ Gas Generating Systems container (BD, 260002) with an anaerobe sachet (BD, 260001) and incubated in a static incubator. Colony counts were determined systematically by first imaging the plates in a gel imager (ChemiDoc XRS+, ImageLab v.6.1, Bio-Rad) using UV transillumination and the standard emission filter (580/120) and then programmatically counting the colonies using Fiji (ImageJ)[31]. Total viable CFU counts per dose were calculated by multiplying the colony counts by the plated dilution factor (to get a CFU per millilitre value) and then the total rehydration volume. Viable CFU counts per dose were then converted to CFU counts per gram by dividing by the mass of the specific dose analysed. A dose includes only the recovered microbial fraction as described above (that is, internal contents of the dosage form).

Total cell counts per dose were determined using an automated microscopic cytometer (QUANTOM Tx Microbial Cell Counter, Logos Biosystems) according to the manufacturer protocols. Specifically, a 10 µl sample of the rehydrated dose used for the viable CFU counts (above) was mixed with 2 µl of a 1:1 mix of total cell staining dye and enhancer (Logos Biosystems, Q13501) on ice; then 8 µl cell loading buffer (Logos Biosystems, Q13501) and 6 µl of the mixture were loaded low numbers of colonies (that is, the three dilutions act as technical replicates). It also allows us to be agnostic to the actual number of countable colonies on any dilution and to assign a robust value to microbial–material combinations that lead to lawns in some dilutions and biological replicates (that is, we can assign a value to a wider range of viabilities than might be possible by using only the countable spots). This analysis was applied to three independent biological replicates for each material–microbial combination.

Correlation analysis (Fig. 2d) of top-performing materials for each organism was carried out by first normalizing the Viability Scores of each specific material and concentration to the maximum Viability Score observed for each organism. This gives a list of materials (at specific concentrations) for each organism with normalized scores that span the full range from 0 to 1. Then any material with a normalized score above 0.75 was defined as a top material for that organism. Based on which organism sets each of these top-performing materials (at specific concentrations) belonged to, they were binned into the 15 possible sets of one-, two-, three- and four-organism overlaps.

Vial-based viability validation of microbial–material stabilizers

Top-performing microbial–material combinations in the high-throughput pipeline (Fig. 2e) were assayed at a larger scale to determine a more accurate relative viability value (CFU per vial) and assess their short-term (1 day) and long-term (30 days) stabilization potential at room temperature. The analysis described below was repeated for seven to eight replicates for the 1 day time point and for four to five replicates for the 30 day time point.

Microbial strains were precultured overnight in 5 ml liquid medium at the appropriate conditions (above). The overnight culture was diluted to an OD_{600} of 0.00225 in 25 ml liquid medium in a flask and cultured for 24 h at the appropriate conditions (above). The cells were harvested by centrifugation, resuspended in cold PBS at a final OD_{600} of 2.25 and used immediately.

To freeze-dry, each microbial cell suspension (100 µl) was mixed with the target material (300 µl) with a micropipette in autoclaved threaded tube vials (Electron Microscopy Sciences, 60304-04) capped with two-prong lyophilization stoppers (Electron Microscopy Sciences, 60304-41) to the first stop. Within 10 min of mixing, the vials were placed in a tray freeze-dryer with shelves precooled to −40 °C and in aluminium StableTemp vial blocks (Cole-Parmer, EW-36600-44). Samples were dried as described above (in 'High-throughput pipeline for microbial–material stabilizers'). After drying, the chamber was backfilled with nitrogen and opened to the atmosphere, and vials were fully stoppered, sealed with screw caps and stored in the dark at 23 °C for the specified time.

The viability of the stored samples was assessed by hydrating the samples with 4 ml cold PBS (to give a 1:10 dilution relative to the original volume), making tenfold serial dilutions, plating 100 µl of each dilution onto the appropriate solid medium in 100-mm-diameter circle Petri dishes with 4.5 mm plating beads (Zymo), incubating at the appropriate conditions and counting colonies on the dilution that gave ~200–1,000 distinct colonies. CFU counts per vial were determined systematically as described above using a gel imager (ChemiDoc XRS+, ImageLab v.6.1, Bio-Rad) and Fiji (in 'Viability survey of commercial probiotic and microbial products').

Validation of two-material combinations

E. coli Nissle 1917 (Supplementary Fig. 6) was prepared as described above (in 'Vial-based viability validation of microbial–material stabilizers') with a few modifications: the flask culture was 2 l and incubated for 17–18 h and the final cell suspension in PBS was set to an OD_{600} of 10–12.

To freeze-dry, the cell suspension (12.5 ml) was mixed with the target material combination (37.5 ml), poured into rectangular one-well plates and placed into a tray freeze-dryer with shelves precooled

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onto haemocytometer slides (Logos Biosystems, Q12001). Slides were centrifuged for 10 min at 300*g* in a slide microcentrifuge (Logos Biosystems, Q10002) and then loaded into the cell counter. Four or five separate doses of each product were prepared as described, and each prepared sample was loaded and quantified in duplicate. For products that gave counts above 1×10^9 cells per millilitre during an initial analysis, the rehydrated dose was first diluted tenfold in PBS before sample preparation. Total cells per dose were calculated by multiplying the cells per millilitre, as reported by the instrument, by the dilution factor (if any) and the total rehydration volume as specified in Supplementary Table 1. Total cell counts per dose were converted to cell counts per gram by dividing by the mass of the dose analysed. A dose includes only the recovered microbial fraction as described above (that is, internal contents of the dosage form).

Percent viable cells relative to promised cells were calculated by dividing the viable CFU counts per gram by the promised cell counts per gram. Percent viable cells relative to total cells were calculated by dividing the total CFU counts per gram by the total cell counts per gram.

Heat stress test of commercial probiotic and microbial products

Three or four doses of each product (Supplementary Fig. 2) as listed in Supplementary Table 1 were incubated at 50 °C for 24 h in their original dosage form in a static incubator. Three or four control doses were kept at the manufacturer recommended storage temperature (4 °C or 23 °C as noted in Supplementary Fig. 2) for the same period. The viable CFU counts per gram for heat-stressed and control doses were determined as described above (in 'Viability survey of commercial probiotic and microbial products'). Percent retention of viability was calculated by dividing each heat-stressed CFU count per gram by the mean of the control CFU counts per gram.

Microbial strains used and general culture conditions

Details of microbial strains used are listed in Supplementary Table 4, and medium components are listed in Supplementary Table 3.

E. coli Nissle 1917 was isolated from the commercial product Mutaflor and routinely cultured on LB agar at 37 °C in a static incubator; or in LB broth in 14 ml culture tubes or in vented, baffled culture flasks shaken at 250 rpm at 37 °C in a shaker incubator.

S. boulardii was isolated from the commercial product Florastor and routinely cultured on yeast extract–peptone–dextrose (YPD) agar at 30 °C in a static incubator or in YPD broth in 14 ml culture tubes or in vented, baffled culture flasks shaken at 250 rpm at 30 °C in a shaker incubator.

E. meliloti Rm1021 was purchased from the ATCC and routinely cultured on tryptone–yeast extract (TY) agar at 30 °C in a static incubator or in TY broth in 14 ml culture tubes or in vented, baffled culture flasks shaken at 250 rpm at 30 °C in a shaker incubator.

L. plantarum NC8 was purchased from the Culture Collection University of Gothenburg and routinely cultured on De Man–Rogosa–Sharpe agar or in De Man–Rogosa–Sharpe broth in sealed 14 ml culture tubes or in sealed culture flasks at 37 °C in a static incubator.

E. coli DH5α was purchased from Thermo Fisher and cultured as described for *E. coli* Nissle 1917.

The pathogens *S. flexneri* 2a, *S. enterica* Infantis and *S. sonnei* were purchased from the ATCC and cultured as described for *E. coli* Nissle 1917.

High-throughput pipeline for microbial–material stabilizers

Microbial strains (Figs. 2a–c, 3a and 5d and Supplementary Figs. 3, 4 and 17) were precultured overnight in 5 ml liquid medium at the appropriate conditions (above). The overnight culture was diluted 1:1,000-fold into three replicates of 125 ml of fresh liquid medium in a flask and cultured for 24 h. The optical density at 600 nm (OD_{600}) of each replicate culture was recorded (typically 2–2.5 for *E. coli* Nissle 1917, *E. meliloti* and *L. plantarum* and 5–6 for *S. boulardii*) and the cells were pelleted at 3,220 relative centrifugal force (rcf) for 15 min. The cell pellets were then resuspended in PBS (*E. meliloti*, *L. plantarum*, *S. boulardii*) or spent medium (*E. coli* Nissle 1917) to a final OD_{600} of 2.25 and used immediately.

The material library was prepared by mixing each material with ultrapure water to the specified concentration (1X or 5X) as listed in Supplementary Table 2. The vendors and product numbers for all materials are listed in Supplementary Table 2. The material solutions were arrayed in sealed, deep well plates and kept at −20 °C until needed. On the day of use, the material plates were thawed at room temperature, vortexed well and centrifuged briefly.

For the two-material combinations with melibiose, the arrayed material library was first mixed 1:1 with melibiose or water as a control at the 1X concentration of all components (Supplementary Table 2). This gives a 0.5X final concentration of each component in Fig. 3a relative to the one-material library results reported in Fig. 2c and Supplementary Fig. 4.

To freeze-dry, each replicate microbial cell suspension (25 µl) was mixed with the arrayed materials (75 µl) in batch into flat-bottom 96-well plates using a liquid handling robot (Tecan, EVO 150). The arrayed microbial–material plates were immediately placed into a tray freeze-dryer (Labconco, FreeZone Stoppering Tray Dryer) with shelves precooled to −40 °C and on custom 0.25 inch aluminium pedestals to ensure heat transfer to the plates. The samples were annealed at −20 °C and then dried at −20 °C and 0.1 mbar for 12 h (nominal) and then at 37 °C for 3 h (nominal). The chamber was backfilled with nitrogen and opened to the atmosphere and then the plates were covered loosely with parafilm, capped and placed in nitrogen-flushed ziplock bags with desiccant (Drierite, granular −8 mesh). The bagged, freeze-dried plates were stored in the dark at the specified temperature (23 °C or 50 °C) for 24 h.

The viability of the stored microbial–material combinations was determined by rehydrating in water, making tenfold serial dilutions in PBS, plating onto the appropriate solid medium and quantifying the resulting colonies. Specifically, using a liquid handling robot (Tecan, EVO 150), each replicate plate was rehydrated with ultrapure water (200 µl per well) and diluted in PBS to make $1{:}10^2$, $1{:}10^3$ and $1{:}10^4$ dilutions (relative to the initial volume before drying), and 4 µl of each dilution was spotted onto solid medium in one-well plates. After incubation at the appropriate culture conditions (above) the plates were imaged in a gel imager (ChemiDoc XRS+, ImageLab v.6.1, Bio-Rad) using UV transillumination and the standard emission filter (580/120).

To assign a Viability Score, using Fiji (ImageJ), the 16-bit Tag Image File Format plate images were programmatically separated into individual spot images, each spot was segmented and several image attributes were measured (raw particle count, mean background value, mean segmented region value and segmented region area). These were used to classify spot images into empty, countable and lawn classes using the ratio of the mean segmented region value to the background value and segmented region area attributes (Supplementary Fig. 3). Raw particle counts for the empty class were set to 0. Raw particle counts for the countable class were interpreted as a true colony count. Raw particle counts for the lawn class were set to a microbe-specific colony count value determined by extrapolating the data observed on a raw count versus a segmented region area plot (Supplementary Fig. 3). For *E. coli* Nissle 1917 and *E. meliloti*, lawns were set to 60 counts. For *S. boulardii*, lawns were set to 50 counts. For *L. plantarum*, lawns were set to 120 counts. These values are sensitive to the colony size at the time of analysis. Finally for each microbial–material combination, we calculated a Viability Score that is the cumulative sum of colony counts at the three dilutions divided by the maximum total possible count (that is, all lawns). The Viability Score thus ranges from 0 to 1. This method allows us to use the data across all three analysed dilutions in order to average out the sampling noise caused from plating and counting to −40 °C and on custom 0.25 inch aluminium pedestals to ensure heat transfer to the plates. Samples were dried as described above (in 'High-throughput pipeline for microbial–material stabilizers'). After drying, the chamber was backfilled with nitrogen and opened to atmosphere, plates were transferred to ziplock bags, the dry material was scrapped into the bag and milled by rolling a plastic tube on the outside of the bag and the resulting powder was kept in the bag with a desiccant pack and stored in the dark at 4 °C until needed.

To prepare samples 'with excipients', the dried bacterial powders were mixed with 1% w/w magnesium stearate, 5% w/w polyvinylpyrrolidone and 64% w/w lactose to give a 30% w/w loading of the bacterial material.

Long-term storage of samples was done by placing ~20 mg (for bacterial powders alone) or ~60 mg (for bacterial powders with excipients) aliquots in 12-well plates, capping the plates and placing them in nitrogen-flushed ziplock bags with desiccant (Drierite, granular −8 mesh). Bagged plates were stored at 37 °C in the dark for the indicated time.

The viability (by pin replicator) of the stored samples was assessed by hydrating the samples with 1 ml PBS, making tenfold serial dilutions, spotting ~1.5 µl of each dilution in triplicate onto LB agar in one-well rectangular plates using a pin replicator (V&P Scientific, VP 407A), incubating at 37 °C and counting colonies on the strongest dilution that gives >1 colony per spot. Total CFU counts per sample were calculated by first dividing spot counts by the spotted volume and multiplying by the dilution factor and total rehydration volume, and then averaging across the three spotting replicates. Total CFU counts per mass were calculated by dividing the CFU counts per sample by the mass of the stored sample. Three independently stored replicate samples were analysed as described. The percent viability at each time point was calculated by dividing the total CFU counts per mass by the mean total CFU counts per mass of three replicate samples at day 0.

Note that this method tends to systematically underestimate the true viable CFU counts but is a robust measure of relative viability. When absolute CFU per milligram values or CFU per millilitre values quantified by this method are reported, they are explicitly labelled 'CFU mg^{-1} (pin replicator)' or 'CFU ml^{-1} (pin replicator)'.

Optimization of two-material formulations

To find optimal concentrations for each of the components in the selected two-material formulations for *E. coli* Nissle 1917 (Fig. 3b), we screened a combinatorial library of component concentrations using the high-throughput pipeline described above (in 'High-throughput pipeline for microbial–material stabilizers') with a few modifications: the flask culture was 2 l and incubated for 17–18 h, the final cell suspension in PBS was set to an OD_{600} of 10–12, dried plates were stored at 37 °C for 23 days, viability was assessed at the 1:10^5 dilution where only empty or countable spots were present and the raw counts were converted directly to CFUs per millilitre by dividing the spot counts by the spotted volume (4 µl) and multiplying by the dilution factor.

The top-performing combinations were named formulation D (1/9X melibiose + 1X yeast extract) and formulation E (1/9X melibiose + 1/125X yeast extract).

Direct viability comparison of synthetic extremophile *E. coli* Nissle 1917 with Mutaflor

Stabilized *E. coli* Nissle 1917 (Fig. 3c and Supplementary Fig. 11) was prepared as described above with either melibiose (1X; in 'Validation of two-material combinations') or formulation D (in 'Optimization of two-material formulations'). For 100X cultures, the 2 l culture was concentrated 100-fold in PBS before mixing with the stabilizers. Mutaflor capsules were kept intact as manufactured during storage.

Samples were stored in nitrogen-flushed ziplock bags with desiccant (Drierite, granular −8 mesh), kept at 23 °C in the dark for the indicated time.

The viability of the stored samples was determined as described above (in 'Viability survey of commercial probiotic and microbial products').

Ultra-long-term assessment of viability

Stabilized *E. coli* Nissle 1917 (Fig. 3d and Supplementary Fig. 7) was prepared as described above (in 'Validation of two-material combinations') with either melibiose, formulation D, formulation E or 5% w/w maltodextrin (16.5–19.5 dextrose equivalent (DE)) as the commercial control. It was determined from the list of inactive ingredients that maltodextrin was the only stabilizer used in the *E. coli* Nissle 1917 product Mutaflor.

Multi-gram samples of the dry powders were stored in 20 ml glass with desiccant and kept in the dark at the specified temperature. Viability was assessed by retrieving a small quantity (~60 mg) of the stored powders and following the procedure described above (in 'Validation of two-material combinations'), but the samples were rehydrated in ultrapure water instead of PBS. In addition to the pin replicator spotting, one sample was also traditionally plated onto 100-mm-diameter circle Petri dishes as described above (in 'Validation of two-material combinations') to determine the scaling factor between the spotting method and the Petri dish plating. When 'CFU mg^{-1} (calibrated stamp)' is reported, we use the scaled values. Percent viability is not impacted by this scaling factor as it is a relative value.

Viability after exposure to wet granulation, tableting, enteric coating, high humidity and high tableting pressure

Stabilized *E. coli* Nissle 1917 (Fig. 4b–d and Supplementary Fig. 13) was prepared as described above (in 'Validation of two-material combinations') with either formulation D or 5% w/w maltodextrin (16.5–19.5 DE) as the commercial control. Dry powders were mixed with the following excipients: 1% w/w magnesium stearate, 5% w/w polyvinylpyrrolidone and 64% w/w lactose (this gives a 30% w/w loading of the bacterial material).

Wet granulation was carried out by adding 600 µl isopropanol to 2 g of the bacterial–excipient mixture while it was being continually stirred with a stand mixer (Sunbeam, B000COC69C) at room temperature. Stirring was allowed to continue for an additional 5 min, and the resulting paste was passed through a benchtop oscillating granulator (ERWEKA, FGS II) with a 1 mm mesh screen. The collected granules were dried at 45 °C for 20 min in a food dehydrator (Magic Mill, MFD-9100) retrofit with a high efficiency particulate air filter (Vornado, MD1-0022) to maintain sterility. Granules were stored in tubes with desiccant at 4 °C until used.

Tableting was carried out in the open air on a tablet press (Natoli, NP-RD10A) with a 5 × 5 mm circular punch and die, set to 8 mm depth and pressed with 17 kN of force (3.4 kN per tablet equal to a pressure of 173 MPa or the indicated pressure). Alternatively, a 1 × 10 mm (Supplementary Fig. 13b), 1 × 2 mm (Supplementary Fig. 13c) or 6 × 2 mm (Supplementary Fig. 13d) circular punch was used, also set to 8 mm depth and pressed with a force appropriate to achieve the indicated tableting pressure. Either the bacterial–excipient mixture was tableted directly ('direct tablet') or the granulated mixture was tableted ('tableted granules'). Tablets were stored in tubes with desiccant at 4 °C until used.

Enteric coating of the tablets was done by spray coating with a solution of 3.6% w/w of EUDRAGIT S 100 (Evonik) in a 1:1 cosolvent of acetone and isopropanol with 0.36% w/w triethyl citrate as a plasticizer. Spray coating was carried at 23 °C via a spray nozzle with a 0.5 mm opening with the tablets in a rotating pan coater (ERWEKA, DKE) over the course of 1 h. Coated tablets were dried at 40 °C for 2 h in a food dehydrator as described above. Coated tablets were stored in tubes with desiccant at 4 °C until used.

Optimized enteric coating was carried out as above, but the tablet drying step was replaced by open-air drying at room temperature for 1 h.

Exposure to 80% relative humidity was achieved by placing the samples on a pedestal inside the gas phase of a sealed glass vial partially filled with a saturated sodium chloride water solution at 37 °C for the specified times. Relative humidity was confirmed with a digital humidity meter (GSP-6, Elitech).

The viability of each of the processed samples was assessed as described above (in 'Validation of two-material combinations') but samples were rehydrated in water instead of PBS, and the CFUs per mass were calculated by dividing either by the mass of the bacterial fraction (that is, 30% of total mass for samples that include excipients) when comparing against samples without excipients (Fig. 4c), or by the mass of the whole sample when all samples contained excipients (Supplementary Figs. 13 and 14). Coated tablets were first cut in half to expose the centre before rehydration.

Compendial quality control and characterization of bacterial tablets

Uncoated and enteric-coated tablets of *E. coli* Nissle 1917 (Fig. 4d and Supplementary Fig. 14) were prepared as described above (in 'Viability after exposure to wet granulation, tableting, enteric coating, high humidity and high tableting pressure') using the 5 × 5 mm punch set to 8 mm depth and a tableting pressure of 160 MPa and coated with the optimized procedure. The quality attributes of these tablets were assessed based on compendial tests (friability, weight variation, disintegration) and non-compendial tests (breaking force, thickness).

Friability[32] was assessed on a friability tester (FT2, SOTAX Corporation). A preweighed sample of tablets weighing 6.5 g (~65 tablets) was de-dusted and placed in the drum of the friabilator. The test was performed at a rotational speed of 25 ± 1 rpm for 4 min. At the end of the test, tablets were de-dusted and accurately weighed and the percentage mass loss was calculated. The test was run once if no tablet fragmentation was observed, and maximum mass loss did not exceed 1%.

Weight variation[33] was calculated after individually weighing twenty randomly selected tablets. For tablets weighing 130 mg or less, the requirements are met when the weights of not more than two tablets differ more than 10% from the average weight and no tablet differs in weight by more than 20%.

Disintegration[34] in simulated gastric fluid (USP, pH 1.2, no enzymes) of the enteric-coated tablets was assessed at 37 ± 2 °C using a disintegration apparatus (DT2, SOTAX Corporation). Six tablets were randomly selected and placed in each tube of the basket, and the test was run for 1 h. If no disintegration occurred, tablets were blotted dry, and viability was assessed as described above (in 'Viability after exposure to wet granulation, tableting, enteric coating, high humidity and high tableting pressure').

Tablet breaking force[35] was measured parallel to the longest axis using a hardness tester (MT50, SOTAX Corporation). The maximum force required to crush six individual tablets was recorded at a test speed of 3 mm s^{-1}.

Tablet thickness for ten individual tablets was measured with an electronic micrometer.

Tuning of release kinetics via matrix formers

E. coli Nissle 1917 (Fig. 4e) was transformed with a plasmid encoding a constitutively expressed pathway for bioluminescence (luxCDABE) via electroporation. Plasmid and strain details are listed in Supplementary Table 4.

Luminescent *E. coli* Nissle 1917 stabilized with formulation D was prepared as described above (in 'Validation of two-material combinations') with cultures supplemented with 100 µg ampicillin ml^{-1}. The resulting bacterial powder was mixed either with the standard excipients (1% w/w magnesium stearate, 5% w/w polyvinylpyrrolidone and 64% w/w lactose) or with the inclusion of a matrix former (50% w/w hydroxypropyl methylcellulose, 1% w/w magnesium stearate, 5% w/w polyvinylpyrrolidone and 14% w/w lactose). In both cases the bacterial material maintains a 30% w/w loading.

A fraction of each of these two powder mixtures was made into tablets as described above for the 'direct tablet' method. Then either one tablet or the equivalent mass of loose powder (with excipients) was placed into 30 ml ultrapure water in a conical bottom tube and incubated for 24 h at 37 °C on an oscillating tube revolver (Thermo Scientific, 88881001) inside a static incubator. At the specified time points, 200 µl samples were taken and arrayed in white-walled, clear-bottom 96-well plates, and the luminescence was quantified on a luminometer (Tecan, Infinite 200, i-control 1.10). Each of the four sample types (standard or matrix-forming excipients in loose or tableted form) were analysed as described in duplicate.

To calculate the percent released luminescence, at each time point the luminescence value of the tableted samples was divided by the mean luminescence value of the corresponding loose powder samples.

qPCR test for efficacy against enteric pathogen *S. flexneri* 2a after mild storage conditions

Dry stabilized *E. coli* Nissle 1917 (Supplementary Fig. 15) was prepared as described above (in 'Validation of two-material combinations') with either formulation D or 5% w/w maltodextrin (16.5–19.5 DE) as the commercial control. Control 'material-only' samples were prepared in the same way but omitting the bacterial cells. Dry powders were stored at 4 °C for six weeks before use. Fresh *S. flexneri*, *E. coli* Nissle 1917 and *E. coli* DH5α were prepared by culturing overnight in LB at 37 °C, pelleting and resuspending in fresh LB at an OD_{600} of 3.

On the day of the experiment, the dry powders were rehydrated in 500 µl deionized water (20 mg powder ml^{-1} water), diluted to 1.5 ml total volume in water and mixed with 1.5 ml of 2X LB. For fresh cell samples (*E. coli* Nissle 1917 and *E. coli* DH5α), 3 µl of the concentrated overnight culture was mixed with 1.5 ml water and then with 1.5 ml of 2X LB to match the rehydrated dry samples. The vehicle control was prepared by simply mixing 1.5 ml water with 1.5 ml of 2X LB.

These cultures were incubated in triplicate at 37 °C for 2 h and then challenged with 3 µl of the concentrated overnight culture of *S. flexneri* and co-cultured for 10 h at 37 °C. To quantify the remaining viable fraction of *S. flexneri*, the co-cultures were diluted 1:100 into fresh LB (3 ml) and outgrown for 12 h at 37 °C, and the total DNA was extracted (Promega Wizard DNA purification kit). Shigella DNA was quantified by qPCR on a Roche LightCycler 96 using the FastStart Essential DNA Green Master Mix (Roche, 06924204001) and the following primers: forward, CCTTTTCCGCGTTCCTTGA; reverse, CGGAATCCGGAGGTATTGC.

The resulting threshold cycle (C_t) values were converted to DNA concentrations using a standard curve of purified *S. flexneri* DNA. For each experiment the DNA concentrations were normalized to the mean of vehicle control replicates, setting it to 100%.

Direct in vitro comparison of synthetic extremophile *E. coli* Nissle 1917 versus Mutaflor against enteric pathogens after storage at high temperatures

Stabilized *E. coli* Nissle 1917 powder (Fig. 5a and Supplementary Fig. 16) was prepared as described above with formulation D (in 'Optimization of two-material formulations'). The internal powder contents of Mutaflor capsules were retrieved and used. Both bacterial powders were incubated at 37 °C for five days in sealed containers with desiccant packs. Pathogenic bacteria *S. flexneri* 2a, *S. sonnei* and *S. enterica* Infantis were prepared fresh by culturing as described above, washed and resuspended in PBS and then used immediately.

On the day of the experiment, each of the stored bacterial powders was rehydrated in deionized water (47 mg ml^{-1}). Then for each pathogen, in a 14 ml round-bottom tube, 1 ml of either the formulation D-stabilized *E. coli* Nissle 1917 suspension (47 mg) or the Mutaflor suspension (47 mg) was mixed with 1 ml of the specified pathogen

(10^8 CFU) and 2 ml of PBS. These bacterial mixtures were prepared in replicate as noted and incubated at 37 °C for 6 h without shaking.

The pathogen load was quantified by diluting the bacterial mixtures 1:10^4 in PBS, plating 100 µl on MacConkey agar plates, incubating at 37 °C and quantifying the resulting colourless colonies that are indicative of the pathogens that do not ferment lactose. Since the plates contained fairly dense mixtures of both *E. coli* Nissle 1917 (pink, Lac+) and pathogen (colourless, Lac−) colonies, the total pathogen colony count per plate was more accurately calculated by quantifying three to four random fields captured on a Leica M165 C microscope and scaling the average CFU count per field by the field to plate area ratio. The final CFU per millilitre value was then calculated by dividing the CFU per plate value by the plated volume and multiplying by the dilution factor. Image analysis was performed blinded to the experimental groups.

To determine whether viable *E. coli* Nissle 1917 cells were necessary to give the observed effects on the pathogen load, the bacterial powders were rehydrated as above, heat inactivated at 95 °C for 30 min and then mixed with pathogens and tested as described above. Since the mixtures contained only viable pathogens, the pathogen load was assessed via the pin replicator method described above (in 'Validation of two-material combinations') using MacConkey agar plates to confirm the absence of viable *E. coli* Nissle 1917.

Direct in vivo comparison of synthetic extremophile *E. coli* Nissle 1917 versus Mutaflor against enteric pathogen *S. flexneri* 2a after storage at elevated temperatures

To assess the performance of the synthetic extremophile in live animals (Fig. 5b), we used a terminal model of intestinal infection in Yorkshire pigs (50–80 kg). This model uses multiple independent segments of the jejunum in a single animal to increase the replicate and experimental group number while minimizing animal use and variability. The Committee on Animal Care at MIT approved of the pig experiments as follows.

Female pigs were obtained from Tufts University (Cummings Veterinary School) and housed in conventional conditions. The day before the experiment, animals were given a liquid diet. Then, the morning feed was withheld; the pigs were sedated with Telazol (tiletamine plus zolazepam, 5 mg kg^{-1}), xylazine (2 mg kg^{-1}) and atropine (0.04 mg kg^{-1}); and a midline laparotomy was performed to access the jejunum. Intestinal clamps (Mayo-Robson) were used to gently segment the jejunum into independent luminal sections without obstructing blood flow.

Bacterial suspensions of the stored powders were prepared as before (in 'Direct in vitro comparison of synthetic extremophile *E. coli* Nissle 1917 versus Mutaflor against enteric pathogens after storage at high temperatures') but rehydrated at 19 mg ml^{-1} immediately before use. The pathogen *S. flexneri* 2a was prepared as described above but resuspended in saline (0.9%) at 5×10^7 CFU ml^{-1} before use. The lumen of each intestinal section was then injected with 2 ml (38 mg) of either the formulation D-stabilized *E. coli* Nissle 1917 suspension or the Mutaflor suspension and then 2 ml of *S. flexneri* 2a (10^8 CFU). The jejunum and clamps were placed back in the peritoneal cavity and the external incision kept closed for 6 h. Then the jejunum was accessed again, and the luminal contents of each section were retrieved by injecting and then aspirating 10 ml of PBS. The pathogen load of the extracted intestinal liquid was quantified as described above on MacConkey agar with the aid of a microscope.

Animals were euthanized at the end of the experiment with sodium pentobarbital (Fatal Plus) at 115–120 mg kg^{-1}. Euthanasia was confirmed by assessing heartrate.

Plant nodulation assay with stabilized *E. meliloti*

E. meliloti cultures (Fig. 5c,d) were prepared as described above (in 'High-throughput pipeline for microbial–material stabilizers') and freeze-dried with the specified stabilizers following the one-well plate method described in 'Validation of two-material combinations'. The bagged, stabilized powders were exposed to 50 °C for 24 h in a static incubator and then kept in the dark at 4 °C until used.

Nodulation of *M. truncatula* A17 was assayed according to a protocol from Sadowsky et al. modified to growth in pouches[36–38]. Specifically, *M. truncatula* A17 seeds (Noble Research Institute) were scarified by submerging in concentrated sulfuric acid for 8–9 min, washed eight times with sterile water, sterilized by submerging in 8.25% sodium hypochlorite for 90 s, washed eight times with sterile water, allowed to imbibe in sterile water for ~1 h, placed on 1% agar plates, sealed with parafilm and foil, vernalized at 4 °C in the dark for five days and germinated at 23 °C in the dark for 20 h. Germinated seeds with root lengths of ~1.5 cm were transferred to sterilized growth pouches (Mega International, CYG) pre-wetted with 5 ml of 1/2X buffered nod medium (BNM; Supplementary Table 3). Five seeds were placed in each pouch. Pouches were placed between 1 inch foam blocks in a domed seed propagator (EarlyGrow, B07HHR5DGN) and incubated in a plant growth chamber (Fisher Scientific, PR505755L) set to a day–night cycle of 25 °C for 16 h and 21 °C for 8 h. The chamber was kept humid with a large tray of water.

Two days after the seeds were transferred to the pouches, the resulting seedlings were each inoculated with 1 ml of the bacterial samples in 1/2X BNM. Specifically, the heat-stressed stabilized *E. meliloti* powders were rehydrated in 1/2X BNM at a common concentration equivalent to 10^8 CFU ml^{-1} of the initial bacterial culture before freeze-drying. Similarly, freshly grown *E. meliloti* cells were pelleted and resuspended in 1/2X BNM to a concentration of 10^7 CFU ml^{-1}.

The pouches were left undisturbed for six days, watered with 5 ml of 1/2X BNM as necessary and then inspected for nodules every other day. On day 12, the number of nodulated seedlings in each pouch was counted. The percent nodulated seedlings was calculated by dividing the number of nodulated seedlings by the total number of seedlings in the pouch.

Viability after exposure to ionizing radiation

Stabilized *E. coli* Nissle 1917 (Fig. 5e) was prepared as described above (in 'High-throughput pipeline for microbial–material stabilizers') with a few modifications: only melibiose (1X) was used as the stabilizer, 32 replicate samples were arrayed on eight 96-well plates (four replicates per plate), and bagged plates were stored at 4 °C until used. On the day of irradiation, a fresh overnight 5 ml culture of *E. coli* Nissle 1917 was arrayed onto the same plates (four replicates per plate) as a control.

The plates were sequentially placed into a Gammacell irradiator (Best Theratronics, 40 Exactor) and exposed to a cobalt-60 source for the time required to achieve the specified ionizing radiation dose (dose rate, 37.12 Gy min^{-1}). One plate was maintained with the other plates but not exposed to any radiation.

The viability of each of the samples was assessed as described above (in 'Validation of two-material combinations') except that samples were rehydrated with 200 µl PBS directly in the treatment plates to give an initial 1:2 dilution for subsequent serial dilution.

Transmission electron microscopy of dry stabilized microbial materials

Stabilized *E. coli* Nissle 1917 (Fig. 3e and Supplementary Fig. 12) was prepared as described above (in 'Validation of two-material combinations') with either formulation D or 5% w/w maltodextrin (16.5–19.5 DE) as the commercial control. Dry powders were kept at 4 °C ('as dried') or incubated at 50 °C for 6.5 h in sealed bags and then kept at 4 °C ('dried + 50 °C').

To fix the cells, 10 mg of each sample was rehydrated in 500 µl deionized water and pelleted for 3.5 min at 21,000 rcf; the bulk supernatant was removed, leaving behind ~20 µl; and the resulting pellet was resuspended in 30 µl of fixative (2.5% w/v paraformaldehyde, 5% w/v glutaraldehyde and 0.06% picric acid in 0.2 M cacodylate buffer). This cell suspension was pelleted for 10 min at 21,000 rcf at room temperature and then transferred to 4 °C to fix overnight.

To prepare the samples for imaging, the fixed cells were washed in 0.1 M cacodylate buffer; postfixed with 1% osmium tetroxide (OsO_4) and 1.5% potassium ferrocyanide ($KFeCN_6$) for 1 h; washed two times in water and one time in maleate buffer; and incubated in 1% uranyl acetate in maleate buffer for 1 h followed by two washes in water and subsequent dehydration in grades of alcohol (10 min each; 50%, 70%, 90%, 2 × 100%). The samples were then put in propyleneoxide for 1 h and infiltrated overnight in a 1:1 mixture of propyleneoxide and Spurr's resin (EMS). The following day, the samples were embedded in Spurr's resin and polymerized at 60 °C for 48 h.

To image the samples, ultrathin sections (about 80 nm) were cut on a Reichert Ultracut-S microtome, picked up onto copper grids, stained with lead citrate and examined in a JEOL 1200EX transmission electron microscope. Images were recorded with an AMT 2k charge-coupled device camera.

To quantify the observed cell classes, three image fields from each sample type containing ~65–95 cells were used. Cells were manually categorized into one of three classes: 'cell debris (no envelope)', 'detached inner membrane' and 'intact envelope' (representative examples of each are shown in Supplementary Fig. 12c). Totals were quantified using the cell counter plugin in Fiji (ImageJ).

Scanning electron microscopy of dry stabilized microbial materials

The various dry microbial materials (Fig. 4b) were imaged in high vacuum mode using the secondary electron detector from the Hitachi FlexSEM TM-1000 II. Low voltage imaging (3 kV) was used to prevent damage from electron bombardment while providing high surface detail. The powders were mounted using double-sided carbon tape (Ted Pella). Using JEOL USA's Smart Coater, gold was deposited for approximately 2.5 min (~12.5 nm gold coating). This conductive coating prevented excessive surface charging artefacts in images. Typical imaging conditions included a spot intensity of 50 (based on the instrument's unitless scale from 1–100) and a working distance of less than 7 mm.

Data analysis

Numerical data were analysed and plotted using Prism 9 (GraphPad). Statistical tests, replicate numbers and errors are indicated for each relevant figure panel.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Data generated or analysed during this study are included in the Supplementary Information. Further data are available from the corresponding author upon request. Source data are provided with this paper.

Code availability

The customized code used for liquid handling and image analysis is available from the corresponding author upon request.

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Author contributions

M.J. and G.T. conceived and designed the research; M.J. designed the high-throughput pipelines; J.L., Z.V., A.K.-S. and Q.C. established the calibration curves for microbial quantification; M.J. and J.L. designed and performed the commercial probiotics survey; M.J. and J.L. performed the initial material stabilizer evaluations and validations; M.J. and E.K. performed the material stabilizer optimization and validations; M.J. and E.K. designed and performed the synthetic extremophile pharmaceutical manufacturing and viability evaluation; M.J. and J.D.B. designed and performed the ionizing radiation experiment; M.J., J.L. and K.B.M. designed, performed and optimized the plant nodulation assay; G.W.L., H.E. and A.D. designed and performed the in vitro pathogen assays; M.J., G.W.L. and N. Fabian designed the in vivo intestinal pathogen model; M.J., G.W.L., N. Fabian, J.J., N. Fitzgerald and A.D. performed the in vivo pathogen assay; M.Y. performed the blinded quantification of pathogen load; C.K. performed the compendial and non-compendial tests on the bacterial tablets; B.M. performed scanning electron microscopy of the microbial materials; M.J., J.L. and E.K. performed formal analysis of the data; M.J. and G.T. wrote the manuscript; M.J., J.L., E.K. and G.T. edited the manuscript; M.J. and G.T. supervised and managed project progress and personnel; and M.J. and G.T. supervised and managed funding of the project.

Competing interests

M.J., G.T., J.L. and E.K. are co-inventors on patent application 18/477,970 (filed 29 September 2023 by MIT), which describes the microbial-stabilizing materials and processes reported here. M.J. and G.T. consult for VitaKey. The other authors declare no competing interests.

Additional information

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Reporting Summary

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Colony plate images for colony counting were collected with Image Lab 6.1 (Bio-Rad). Spectrometry data (fluorescence, absorbance, luminescence) were collected with i-control 1.10 (Tecan).
Data analysis	GraphPad Prism 10.1.1 (Graphing software, http://www.graphpad.com), images were analyzed with ImageJ2 2.3.0 (Fiji).

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All data needed to evaluate the conclusions in the paper are present in the paper or the Supplementary Materials.

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes for all in vitro measurements were chosen to be n=3-5 based on preliminary experiments on control or baseline conditions. Sample sizes for pig experiments were chosen to be n=5 intestinal segments based on preliminary experiments on control or baseline conditions. Sample sizes for plant experiments were chosen to be n=3-8 based on preliminary experiments on control or baseline conditions
Data exclusions	No data was excluded from analysis.
Replication	Experiments were performed with three or more independent replicates. Key samples from high throughput screens were re-evaluated in larger formats using alternate quantification methods. All replication attempts were successful.
Randomization	All samples and animals were randomly assigned to each experimental group.
Blinding	Blinding is not relevant to the study. The data presented does not require subjective judgment or interpretation. Specifically, the main data type in the study was colony counts that were performed programmatically in a standardized manner across all experimental groups as described in the methods, which reduces unintended experimenter bias from the analysis being performed unblinded

Reporting for specific materials, systems and methods

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Materials & experimental systems

n/a	Involved in the study	
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☒	☐	Eukaryotic cell lines
☒	☐	Palaeontology and archaeology
☐	☒	Animals and other organisms
☒	☐	Clinical data
☒	☐	Dual use research of concern

Methods

n/a	Involved in the study	
☒	☐	ChIP-seq
☒	☐	Flow cytometry
☒	☐	MRI-based neuroimaging

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	Female Yorkshire pigs (50-95kg, 6-7 months at time of experiment, terminal), were received from Cummings Veterinary School at Tufts University in Grafton, MA.
Wild animals	No wild animals were used.
Reporting on sex	For pig experiments, female pigs were used. We use predominantly female swine because they are more amenable to being socially housed and tend to be less aggressive.
Field-collected samples	No field samples were used.
Ethics oversight	Approval for the pig experiments was obtained from the Committee on Animal Care at the Massachusetts Institute of Technology.

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